

Complete PostgreSQL Commands Documentation

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1 PostgreSQL Shell Commands

1.1 Start PostgreSQL Shell

```
sudo -u postgres psql
```

Starts PostgreSQL interactive terminal.

1.2 Login Using Specific User

```
psql -U priyanshu
```

Logs in using a PostgreSQL user.

1.3 Connect to Specific Database

```
psql -U priyanshu -d fastAPI_db
```

Connects to a specific database.

1.4 Exit PostgreSQL

```
\q
```

Exits PostgreSQL shell.

2 Database Commands

2.1 Show All Databases

```
\l
```

Lists all databases.

2.2 Create Database

```
CREATE DATABASE fastAPI_db;
```

Creates a new database.

2.3 Delete Database

```
DROP DATABASE fastAPI_db;
```

Deletes a database.

2.4 Rename Database

```
ALTER DATABASE fastAPI_db  
RENAME TO newdb;
```

Renames database.

2.5 Connect to Database

```
\c fastAPI_db
```

Connects to a database.

3 User and Role Commands

3.1 Show All Users / Roles

```
\du
```

Displays all PostgreSQL users and roles.

3.2 Create User

```
CREATE USER priyanshu  
WITH PASSWORD '1234';
```

Creates new PostgreSQL user.

3.3 Create Superuser

```
CREATE USER priyanshu  
WITH SUPERUSER  
PASSWORD '1234';
```

Creates user with admin privileges.

3.4 Change Password

```
ALTER USER priyanshu  
WITH PASSWORD 'newpassword';
```

Changes user password.

3.5 Rename User

```
ALTER USER oldname  
RENAME TO newname;
```

Renames PostgreSQL user.

3.6 Delete User

```
DROP USER priyanshu;
```

Deletes PostgreSQL user.

3.7 Grant Superuser Access

```
ALTER USER priyanshu  
WITH SUPERUSER;
```

Grants admin privileges.

3.8 Remove Superuser Access

```
ALTER USER priyanshu  
WITH NOSUPERUSER;
```

Removes admin privileges.

4 Permission Commands

4.1 Grant Database Permissions

```
GRANT ALL PRIVILEGES  
ON DATABASE fastAPI_db  
TO priyanshu;
```

Gives all permissions on database.

4.2 Revoke Permissions

```
REVOKE ALL PRIVILEGES  
ON DATABASE fastAPI_db  
FROM priyanshu;
```

Removes permissions.

5 Table Commands

5.1 Show Tables

```
\dt
```

Displays all tables.

5.2 Describe Table

```
\d books
```

Shows table structure.

5.3 Detailed Table Info

```
\d+ books
```

Shows detailed table information.

5.4 Create Table

```
CREATE TABLE books(  
id SERIAL PRIMARY KEY,  
title VARCHAR(100)  
);
```

Creates table.

5.5 Delete Table

```
DROP TABLE books;
```

Deletes table.

5.6 Rename Table

```
ALTER TABLE books  
RENAME TO products;
```

Renames table.

5.7 Add Column

```
ALTER TABLE books  
ADD COLUMN author VARCHAR(100);
```

Adds new column.

5.8 Remove Column

```
ALTER TABLE books  
DROP COLUMN author;
```

Deletes column.

6 CRUD Operations

6.1 Insert Data

```
INSERT INTO books(title)
VALUES('Atomic_Habits');
```

Inserts data.

6.2 Read Data

```
SELECT * FROM books;
```

Reads all rows.

6.3 Update Data

```
UPDATE books
SET title='Deep_Work'
WHERE id=1;
```

Updates data.

6.4 Delete Data

```
DELETE FROM books
WHERE id=1;
```

Deletes rows.

7 Schema Commands

7.1 Show Schemas

```
\dn
```

Displays all schemas.

7.2 Create Schema

```
CREATE SCHEMA test;
```

Creates schema.

8 Utility Commands

8.1 Show Current Database

```
SELECT current_database();
```

Shows connected database.

8.2 Show Current User

```
SELECT current_user;
```

Shows current logged-in user.

8.3 Show PostgreSQL Version

```
SELECT version();
```

Displays PostgreSQL version.

8.4 Show Connection Info

```
\conninfo
```

Displays current connection details.

8.5 Show Config File

```
SHOW config_file;
```

Shows PostgreSQL config location.

8.6 Show Authentication File

```
SHOW hba_file;
```

Shows *pg_hba.conf* location.

9 Linux Service Commands

9.1 Start PostgreSQL Service

```
sudo systemctl start postgresql
```

Starts PostgreSQL service.

9.2 Stop PostgreSQL Service

```
sudo systemctl stop postgresql
```

Stops PostgreSQL service.

9.3 Restart PostgreSQL Service

```
sudo systemctl restart postgresql
```

Restarts PostgreSQL service.

9.4 Check PostgreSQL Status

```
sudo systemctl status postgresql
```

Shows PostgreSQL service status.

9.5 Check PostgreSQL Port

```
sudo ss -ltnp | grep 5432
```

Checks running PostgreSQL port.

10 Connection URL

10.1 Connection Format

```
postgresql://USER:PASSWORD@HOST:PORT/DATABASE
```

General PostgreSQL connection format.

10.2 Example Connection URL

```
postgresql://priyanshu:1234@localhost:5432/fastAPI_db
```

Example connection string.

11 Common Errors and Fixes

11.1 Role Does Not Exist

```
FATAL: role "priyanshu" does not exist
```

Fix:

```
CREATE USER priyanshu  
WITH PASSWORD '1234';
```

11.2 Database Does Not Exist

```
FATAL: database "fastAPI_db" does not exist
```

Fix:

```
CREATE DATABASE fastAPI_db;
```

11.3 Password Authentication Failed

```
FATAL: password authentication failed
```

Fix:

```
ALTER USER priyanshu  
WITH PASSWORD 'newpassword';
```

11.4 Permission Denied

```
permission denied
```

Fix:

```
GRANT ALL PRIVILEGES  
ON DATABASE fastAPI_db  
TO priyanshu;
```

12 Recommended Beginner Setup

```
CREATE USER priyanshu  
WITH PASSWORD '1234';  
  
CREATE DATABASE fastAPI_db;  
  
GRANT ALL PRIVILEGES  
ON DATABASE fastAPI_db  
TO priyanshu;
```